Quantitative Methods for Management

Tutorial Exercise Sheet: Sensitivity and Parametric Analysis

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Recap: Sensitivity Analysis. Sensitivity analysis examines whether an optimal basis remains optimal when the data of a linear program change. To perform the sensitivity analysis, we first need to know what the changes influence. To identify this, we can use tools from the simplex algorithm. Denote as $\mathbf{A}$ the constraint matrix including the slack columns. Similarly, $\mathbf{x}$ includes the slack variables, and their coefficients in $\mathbf{c}$ are zero.

Denote as B the indices of the basic variables and N those of the nonbasic variables. The columns indexed by B form a square, invertible matrix $\mathbf{A}_B$; the remaining columns form $\mathbf{A}_N$. Setting $\mathbf{x}_N = 0$ gives the basic solution $\mathbf{x}_B = \mathbf{A}_B^{-1}\mathbf{b}$. The two conditions to check are

$$\begin{aligned} \mathbf{A}_B^{-1}\mathbf{b} \geq 0 \quad &\text{(feasibility)}, \\ \mathbf{r}_N^T = \mathbf{c}_N^T - \mathbf{c}_B^T\mathbf{A}_B^{-1}\mathbf{A}_N \leq 0 \quad &\text{(optimality for maximization)}. \end{aligned}$$

All inequalities hold componentwise. The first condition ensures that the basic variables are nonnegative. The second ensures that increasing a nonbasic variable from zero cannot improve the objective. If both conditions hold, the basis is optimal. For a minimization problem, the reduced-cost condition is instead $\mathbf{r}_N \geq 0$.

The reduced costs also follow from the dual variables (shadow prices):

$$\mathbf{u}^* = (\mathbf{A}_B^T)^{-1}\mathbf{c}_B, \qquad \mathbf{r} = \mathbf{c} - \mathbf{A}^T\mathbf{u}^*, \qquad \mathbf{r}_B = 0.$$

Thus, a reduced cost compares a variable's objective coefficient with the value of the resources it uses. Include the slack variables when checking the reduced-cost condition.

Starting from an optimal basis, consider each of the following changes separately:

- **Only c changes: check optimality.** The basic solution is unchanged and remains feasible. Recompute the reduced costs to determine whether the same basis is still optimal.
- **Only b changes: check feasibility.** The reduced costs are unchanged. Recompute $\mathbf{x}_B = \mathbf{A}_B^{-1}\mathbf{b}$; the basis remains optimal as long as all basic variables remain nonnegative. The production quantities may change even though the basis stays the same.
- **Only $\mathbf{A}_N$ changes: check optimality.** Since the nonbasic variables are zero, the basic solution is unchanged. Recompute the reduced costs using the new nonbasic columns.
- **Coefficients in $\mathbf{A}_B$ change: check both conditions.** First verify that $\mathbf{A}_B$ is still invertible; otherwise, these columns no longer form a basis. Then recompute both the basic solution and the reduced costs.

If a check fails, the original basis no longer satisfies both conditions, and the changed problem requires further analysis.

Exercise 1 (Sensitivity Analysis)**.** A company produces three types of pasta: spaghetti, penne, and fusilli. Each week, it has 100 hours of shaping machine time and 80 hours of drying equipment time available. Producing 100 kg of spaghetti requires 2 hours of shaping and 1 hour of drying, and generates a net profit of 400 €. Producing 100 kg of penne requires 1 hour of shaping and 2 hours of drying, and generates a net profit of 500 €. Producing 100 kg of fusilli requires 1 hour of shaping and 1 hour of drying, and generates a net profit of 250 €. The company aims to maximize their profit.

(a) Define the decision variables and formulate the company's production problem as a linear program. Proceed to formulate the dual problem and solve it. Determine an optimal production plan and the maximum weekly profit. Interpret the optimal dual variables.

(b) Solve the primal problem again, this time using the simplex algorithm. Compare the result with Part (a).

(c) Suppose the available shaping time increases from 100 to 130 hours per week. Does the original optimal basis remain optimal? Determine the new production plan and weekly profit. Relate the profit increase to the shadow price of shaping capacity.

(d) Suppose the net profit from spaghetti decreases from 400 € to 350 € per 100 kg. Does the original production plan remain optimal? Determine the resulting maximum weekly profit.

(e) Suppose the net profit from fusilli increases from 250 € to 350 € per 100 kg. Use the original dual solution to check whether the original production plan remains optimal. Explain whether the company should now produce fusilli.

Analyze the changes in Parts (c)–(e) separately, starting from the original program.

Recap: Parametric Analysis. Parametric analysis determines how the optimal solution and objective value depend on a parameter. When only the objective coefficients depend on a parameter ρ, the feasible region stays fixed and does not depend on the parameter. If this region is nonempty and bounded, as in the following exercise, we can proceed as follows:

- **Compute the vertices of the feasible region.** In two dimensions, find the intersections of pairs of constraint boundary lines, including the nonnegativity boundaries. Keep only the points that satisfy all constraints. Denote the distinct vertices by $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(k)}$.
- **Compare the objective values at all vertices.** Write the value at vertex i as $z_i(\rho) = \mathbf{c}(\rho)^T \mathbf{x}^{(i)}$. A vertex $\mathbf{x}^{(i)}$ is optimal if and only if

$$\begin{aligned} z_i(\rho) &\geq z_j(\rho) \quad \text{for every } j \quad \text{(maximization)}, \\ z_i(\rho) &\leq z_j(\rho) \quad \text{for every } j \quad \text{(minimization)}. \end{aligned}$$

  For each vertex, solve these inequalities together with the given restrictions on ρ to find the parameter range in which it is optimal.

The optimal objective value is the largest of the functions $z_i(\rho)$ for maximization, or the smallest for minimization. Use the parameter ranges to write it as a piecewise-defined function. At a parameter value where two vertices tie for the optimal value, every point on the line segment joining them is also optimal.

If only the RHS $\mathbf{b} = \mathbf{b}(\rho)$ depends on the parameter, the primal feasible region changes with different values of the parameter ρ. However, we can work with the dual:

$$\begin{array}{rl} \max\limits_{\mathbf{x}} & \mathbf{c}^T\mathbf{x} \\ \text{subject to} & \mathbf{A}\mathbf{x} \le \mathbf{b}(\rho), \\ & \mathbf{x} \ge 0, \end{array} \quad \longrightarrow \quad \begin{array}{rl} \min\limits_{\mathbf{u}} & \mathbf{b}(\rho)^T\mathbf{u} \\ \text{subject to} & \mathbf{A}^T\mathbf{u} \ge \mathbf{c}, \\ & \mathbf{u} \ge 0. \end{array}$$

The dual feasible region is fixed because $\mathbf{A}$ and $\mathbf{c}$ do not change with different values of ρ. The parameter now appears only in the dual objective coefficients, so we can use the vertex comparisons as above:

- Compute the dual vertices $\mathbf{u}^{(1)}, \ldots, \mathbf{u}^{(k)}$ and their objective values $w_i(\rho) = \mathbf{b}(\rho)^T\mathbf{u}^{(i)}$.
- Since the dual is a minimization problem, solve $w_i(\rho) \le w_j(\rho)$ for every j to find the range where vertex i is optimal.

Apply these comparisons for parameter values where the primal is feasible and has a finite optimal value. Strong duality then gives the primal optimal value as $z(\rho) = \min_i w_i(\rho)$. The dual region may be unbounded; a finite minimum is still attained at a vertex for the dual above.

To recover an optimal primal solution, use the optimality conditions with the selected dual optimum $\mathbf{u}^*(\rho)$. If $u_i^*(\rho) > 0$, resource constraint i binds. If $(\mathbf{A}^T\mathbf{u}^*(\rho))_j > c_j$, set $x_j = 0$. Solve the resulting equations and check all primal constraints, including nonnegativity. This gives the optimal quantities as functions of the parameter.

Exercise 2 (Parametric Analysis of an Objective Coefficient)**.** For the real parameter $\rho \ge 0$, consider the linear programming problem

$$\begin{array}{rl} \max\limits_{\mathbf{x}} & \rho x_1 + 2x_2 \\ \text{subject to} & 4x_1 + 3x_2 \le 22, \\ & 2x_1 + x_2 \le 10, \\ & 2x_1 + 5x_2 \le 10, \\ & x_1, x_2 \ge 0. \end{array}$$

(a) Determine the optimal solutions as a function of ρ. Write down the problem in vector-matrix notation and check if any constraints are redundant.

(b) Write the optimal objective value $z(\rho)$ as a piecewise-defined function and sketch its graph.

(c) Determine the optimal solution when $\rho = 1$. Find the interval of parameter values for which this solution remains optimal.

Exercise 3 (Parametric Analysis)**.** A company produces both orange and apple juice. Producing 100 liters of orange juice requires 2 hours of blending and 1 hour of bottling, and generates a net profit of 40 €. Producing 100 liters of apple juice requires 1 hour of

blending and 1 hour of bottling, and generates a net profit of 30 €. The company has 16 hours of blending time and 10 hours of bottling time available per day. The company aims to maximize their profit.

(a) Define the decision variables and formulate the company's production problem as a linear programming problem. Solve the problem graphically. Determine the optimal production plan, maximum daily profit, and binding constraints.

For the following exercises you may use the solution of Part (a):

$$\mathbf{c} = \begin{pmatrix} 40 \\ 30 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 16 \\ 10 \end{pmatrix}, \quad \mathbf{x}^* = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

(b) Suppose the available blending time is now t hours per day, where $t \geq 0$, while bottling time remains fixed at 10 hours. Formulate and solve the dual problem as a function of t. Use complementary slackness to determine the optimal production plan and identify the parameter values at which the binding constraints change.

(c) Use the dual solution to write the maximum daily profit $V(t)$ as a piecewise-defined function and sketch its graph. Interpret the slope on each interval using shadow prices.

(d) Use the shadow prices to determine the smallest blending capacity beyond which additional blending time does not increase profit. Explain which resource limits production in this range.

(e) Using the optimal shadow-price vector $\mathbf{u}^*(t)$ from Part (b), compute the reduced cost vector $\mathbf{r}(t) = \mathbf{c} - \mathbf{A}^T\mathbf{u}^*(t)$. Use it together with complementary slackness to recover the optimal production plan in each parameter range without applying the simplex algorithm.